Project Name: Project 1: Voting System **Team#1**

Test Stage: Unit ✓ System __

Test Date: 11/3/25

Test Case ID#: TC_001

Name(s) of Testers: Adam Banini

Test Description: Test checks if assignLiveBallots method adds all live ballots to the correct candidate as indicated by the Ballot object's preference.

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

STVTest.java
testAssignLiveBallots()

Automated: yes ✓ no ___

Results: Pass ✓ Fail

Preconditions for Test: STV object has valid Ballots in liveBallots, STV object has valid Candidates

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Assign live ballots	Ballot with a valid candidate as the primary choice	Adds ballot to primary choice candidate	Java assertion of expected results passed	
2	Assign dead ballots	Ballot with no valid candidates as choice	Add ballot to dead ballots	Java assertion of expected results passed	
3	Assign ballot with 1st choice not valid	Ballot with valid 2nd choice	Add ballot to 2nd choice candidate	Java assertion of expected results passed	
4	Assign ballot with 1st and 2nd choice not valid	Ballot with valid 3rd choice	Add ballot to 3rd choice candidate	Java assertion of expected results passed	
5					

Post condition(s) for Test: All Ballots are added into their preferred candidate's ballot list.

Project Name: Project 1: Voting System **Team#1**

Test Stage: Unit ✓ System __ **Test Date: 11/3/25**

Test Case ID#: TC_002 **Name(s) of Testers: Adam Banini**

Test Description: Test checks if chooseLoser returns the index of the candidate that should be the loser at the end of a given round.

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

STVTest.java
testChooseLoser()

Automated: yes ✓ no

Results: Pass ✓ Fail

Preconditions for Test: STV object has valid Candidates

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Clear loser	One candidate at the end of the round has the least votes	The clear loser is added to the losers list	Java assertion of expected results passed	
2	Tie with zero votes	Two candidates tied at the end of a round with 0 votes	One of the two losers is chosen randomly and added to losers	Java assertion of expected results passed	
3	Tie with votes	Two candidates tied at the end of a round with votes	Candidate with the "worst" best ballot is made a loser	Java assertion of expected results passed	Best ballot = lowest shuffle ID Worst ballot = highest shuffle ID

Post condition(s) for Test: An integer correlating to the index of the losing Candidate stored in STV's candidate list.

Project Name: Project 1: Voting System **Team#1**

Test Stage: Unit ✓ System __

Test Date: 11/3/25

Test Case ID#: TC_003

Name(s) of Testers: Adam Banini

Test Description: Test checks if breakTie returns the index of a single candidate determined to be a loser.

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

STVTest.java
testBreakTie()

Automated: yes ✓ no ___

Results: Pass ✓ Fail ___

Preconditions for Test: STV object has valid Candidates, Candidates delivered in the tied list are tied

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	One tied candidate	One candidate at the end of the round has the least votes	The clear loser is added to the losers list	Java assertion of expected results passed	
2	Two tied candidates with 0 votes	Two candidates tied at the end of a round with 0 votes	One of the two losers is chosen randomly and added to losers	Java assertion of expected results passed	
3	Two tied candidates with votes	Two candidates tied at the	Candidate with the "worst" best	Java assertion of expected results passed	Best ballot = lowest shuffle ID

		end of a round with votes	ballot is made a loser		Worst ballot = highest shuffle ID
4					

Post condition(s) for Test: An integer correlating to the index of the losing Candidate stored in STV's candidate list.

Project Name: Project 1: Voting System **Team#1**

Test Stage: Unit ✓ System __

Test Date: 11/3/25

Test Case ID#: TC_004

Name(s) of Testers: Adam Banini

Test Description: Test checks if promoteToFill fills the winners list with the most recent losers in the reverse order in which they lost.

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

STVTest.java
testPromoteToFill()

Automated: yes ✓ no ___

Results: Pass ✓ Fail

Preconditions for Test: STV object has valid Candidates, All Candidates are either in the losers or winners list

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	One seat must be filled	STV election past round stage with one open seat and two losers	The most recent loser is added to the winners list	The most recent loser assertion passed	
2	Three seats must be filled	STV election past round stage with three open seats and four losers	The three most recent losers are added to the winners list in order	The three most recent losers assertion passed	

Post condition(s) for Test: Winner list size is equal to totalSeats, Winners list matches the expected winners list of an STV election

Project Name: Project 1: Voting System **Team#1**

Test Stage: Unit ✓ System __

Test Date: 11/3/25

Test Case ID#: TC_005

Name(s) of Testers: Adam Banini

Test Description: Test checks if calculateDroop returns the correct integer to represent the droop quota of the election

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

STVTest.java
testCalculateDroop()

Automated: yes ✓ no

Results: Pass __✓__ Fail_____

Preconditions for Test: STV object has non-null totalSeats and liveBallots

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	One seat 5 ballots	STV election with one seat and 5 ballots	Droop is set to be 3	Java assertion of expected result passed	
2	2 seats 4 ballots	STV election with 2 seats and 6 ballots	Droop is set to be 2	Java assertion of expected results passed	
3	3 seats 12 ballots	STV election with 3 seat and 8 ballots	Droop is set to be 4	Java assertion of expected results passed	
4					

Post condition(s) for Test: Droop quota is returned as an integer.

Project Name: Project 1: Voting System **Team#1**

Test Stage: Unit ✓ System __ **Test Date: 11/3/25**

Test Case ID#: TC_006 **Name(s) of Testers: Adam Banini**

Test Description: Test checks if isFinished correctly determines if the round stage of STV is complete i.e. all candidates are winners or losers.

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

STVTest.java
testIsFinished()

Automated: yes ✓ no

Results: Pass ✓ Fail

Preconditions for Test: STV object has valid Candidates

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	All winners, no losers	STV where all candidates are winners and none are losers	True	Java assertion of expected results passed	
2	No winners, all losers	STV where no candidates are winners and all are losers	True	Java assertion of expected results passed	
3	No winners, no losers	STV where no candidates are assigned	False	Java assertion of expected results passed	
4	Some winners, some losers, some nothing	STV where some candidates are winners and some are losers, and some are not assigned	False	Java assertion of expected results passed	
5	Some winners, rest losers	STV where some candidates are winners and rest are losers	True	Java assertion of expected results passed	

Post condition(s) for Test: Boolean correctly indicating if the round stage is finished.

Test Stage: Unit ✓ System __

Test Date: 11/3/25

Test Case ID#: TC_007

Name(s) of Testers: Adam Banini

Test Description: Test checks if runElection prints the proper winners

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

STVTest.java
testRunElection()

Automated: yes ✓ no __

Results: Pass ✓ Fail

Preconditions for Test: STV object has valid Candidates and Ballots and totalSeats is indicated.

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Clear Winners	STV election with ballots indicating Alice as a clear winner	Alice is printed to the screen	Alice was printed as a winner	
2	Tie with 0 vote tie	STV election with a tie where two candidates are tied at 0 votes	A random winner is printed to the screen	Random winner is one of the tied candidates	
3	Tie with 1+ vote tie	STV election with a tie where two candidates are tied at more than zero votes, but Alice is the expected winner	A winner is printed that is always the same as long as shuffle is disabled	Alice was printed as a winner	

Post condition(s) for Test: An integer correlating to the index of the losing Candidate stored in STV's candidate list.

Project Name: Project 1: Voting System **Team#1**

Test Stage: Unit ✓ System __ **Test Date: 11/3/25**

Test Case ID#: TC_008 **Name(s) of Testers: Adam Banini**

Test Description: Test checks if shuffle sets the shuffleId of all ballots to a truly shuffled ordering

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

STVTest.java
testShuffle()

Automated: yes ✓ no ___

Results: Pass ✓ Fail _____

Preconditions for Test: STV object has valid Ballots

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	15 ballots	15 ballots	Majority of shuffl IDs don't match ballot IDs	Java assertion that < 4 ballotID == shuffleID passed	
2					
3					
4					

Post condition(s) for Test: In a group of 15 ballots no more than 4 will have the same ballotId as they do shuffleId

Project Name: Project 1: Voting System **Team#1**

Test Stage: Unit ✓ System __ **Test Date: 11/3/25**

Test Case ID#: TC_009 **Name(s) of Testers: Adam Banini**

Test Description: Test checks if printResults prints the results of the election

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

STVTest.java
testPrintResults()

Automated: yes ___ no ✓

Results: Pass ✓ Fail ___

Preconditions for Test: STV object exists

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	2 winners 2 losers 2 ballots	STV election with 2 winners 2 losers 2 ballots	Prints to screen indicating that there were 2 winners, 2 losers, 2 ballots, 4 candidates, 2 seats	STV Election Results ---------- Total Ballots: 2 Total Seats: 2 Total Candidates: 4 Winners: Alice Bob Losers: Charlie Diana	

2					
3					
4					

Post condition(s) for Test: All of the relevant information contained in the STV object is displayed to the screen

Project Name: Project 1: Voting System **Team#1**

Test Stage: Unit ✓ System __

Test Date: 11/3/25

Test Case ID#: TC_010

Name(s) of Testers: Adam Banini

Test Description: Test checks if generateAuditFile creates a uniquely named txt file with the results of the election

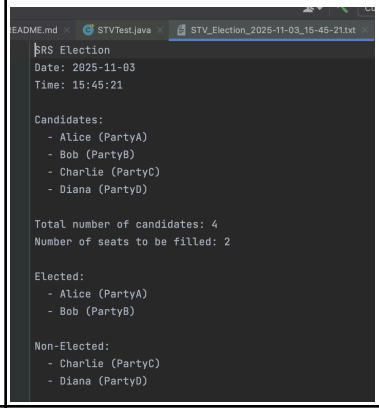
Indicate where are you storing the tests (what file) and the name of the method/functions being used.

STVTest.java
testGenerateAudit()

Automated: yes no ✓

Results: Pass ✓ Fail

Preconditions for Test: STV object exists

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	2 winners 2 losers 2 ballots	STV election with 2 winners 2 losers 2 ballots	Creates an audit file indicating that there were 2 winners, 2 losers, 4 candidates, 2 seats, and the date/time it was generated		

Post condition(s) for Test: All of the relevant information contained in the STV object is written into the txt file

Project Name: Project 1: Voting System **Team# 1**

Test Stage: Unit x_ System __ **Test Date: 11/1/2025**

Test Case ID#: TC_11 **Name(s) of Testers: Rahim Bardai**

Test Description:

Test that the constructor creates an empty File Handler

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes_x__ no ___

Results: Pass x Fail

Preconditions for Test: FileHandler exists, getNumVotes(), getCandidates(), and getBallots() exists, no prior state of data required.

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1					
2	Create new filehandler object	File Handler Class	fileHandler()	fileHandler()	Pass
3	Check none for getNumVotes()	Null	0	0	Pass
4	Check getCandidates() is empty	Null	True	True	Pass
	Check getBallots() is empty	Null	True	True	Pass

Post condition(s) for Test:

The object's variables are set to 0/empty for future use.

Project Name: Project 1: Voting System **Team# 1**

Test Stage: Unit x_ System __

Test Date: 11/1/2025

Test Case ID#: TC_12

Name(s) of Testers: Rahim Bardai

Test Description:

Test that checks if getCandidates() returns the right number of names and order

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes x no

Results: Pass __x___ Fail_____

Preconditions for Test: FileHandler exists, getNumVotes(), getCandidates(), and getBallots() exists, no prior state of data required.

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1					
2	Create tempFile	tempFile	File is created	File is created	Pass

3	Write in sample test cases	Arish, Rahim, Hlina
		
		1,2,3
		2,1,3
			`writer.println("Arish,Rahim,Hlina");`
			
			`writer.println("1,2,3");`
			
			`writer.println("2,1,3");`
				`writer.println("Arish,Rahim,Hlina");`
				
				`writer.println("1,2,3");`
				
				`writer.println("2,1,3");`
					Pass

4	Check getCandidates() array at each index	getCandidates() array	Arish Rahim Hlina	Arish Rahim Hlina	Pass

Post condition(s) for Test:

Project Name: Project 1: Voting System **Team# 1**

Test Stage: Unit x_ System __

Test Date: 11/1/2025

Test Case ID#: TC_13

Name(s) of Testers: Rahim Bardai

Test Description:

Test that checks if getBallots() returns the right number of voting order

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes_x__ no ___

Results: Pass __x___ Fail_____

Preconditions for Test: FileHandler exists, getNumVotes(), getCandidates(), and getBallots() exists, no prior state of data required.

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1					
2	Create tempFile	tempFile	File is created	File is created	Pass
3	Write in sample test cases	Arish, Rahim, Hlina 1,2,3 2,1,3	`writer.println("Arish,Rahim,Hlina");` `writer.println("1,2,3"`	`writer.println("Arish,Rahim,Hlina");` `writer.println("1,2,3");`	Pass

			); writer.println("2,1,3");	writer.println("2,1,3");	
4	Check getCandidates() array at each index	getBallots() array	1 2 3 2 1 3	1 2 3 2 1 3	Pass

Post condition(s) for Test:

Project Name: Project 1: Voting System **Team# 1**

Test Stage: Unit x_ System __

Test Date: 11/1/2025

Test Case ID#: TC_14

Name(s) of Testers: Rahim Bardai

Test Description:

Test that checks if the reader skips empty lines in the ballot file

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes_x__ no ___

Results: Pass x Fail

Preconditions for Test: FileHandler exists, getNumVotes(), getCandidates(), and getBallots() exists, no prior state of data required.

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1					
2	Create tempFile	tempFile	File is created	File is created	Pass
3	Write in sample test cases	Arish, Rahim, Hlina 1,2,3 2,1,3	`writer.println("Arish, Rahim,Hlina");`	`writer.println("Arish,Rahim,Hlina");`	Pass

			writer.println("1,2,3"); writer.println("2,1,3");	writer.println("1,2,3"); writer.println("2,1,3");	
4	Check getCandidates() array at each index	getCandidates() array size	3	3	Pass
	Check the num of votes	getNumVotes()	2	2	Pass

Post condition(s) for Test:

Project Name: Project 1: Voting System **Team# 1**

Test Stage: Unit _x_ System __ **Test Date: November 2, 2025**

Test Case ID#: TC_15 **Name(s) of Testers: Hlina**

Test Description: Test that verifies that output of HelpWindow ExplainFileEntry method prints out the correct message to the screen

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes___ ___

Results: x_____ Passed

Preconditions for Test:
Junit 5 testframe work available

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1					

2	HelpWindow explainFileEntry method is called to output explanation of correct file entry	HelpWindow Class	HelpWindow.explainFileEntry()	HelpWindow.explainFileEntry()	Pass
3					
4					

Post condition(s) for Test:

Explanation of how to enter files into the system has been printed out to the command line user interface.

Project Name: Project 1: Voting System **Team# 1**

Test Stage: Unit _x_ System __

Test Date: November 2, 2025

Test Case ID#: TC_15

Name(s) of Testers: Hlina

Test Description: Test that verifies that output of HelpWindow ExplainVote method prints out the correct message to the screen

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes___ ___

Results: x_____ Passed

Preconditions for Test:
Junit 5 testframe work available

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1					
2	HelpWindow explainVote method is called to output explanation of the voting algorithms offered by the system and to prompt user to select one.	HelpWindow Class	HelpWindow.explainVote()	HelpWindow.explainVote()	Pass
3					
4					

Post condition(s) for Test:
Explanation of voting algorithms offered by the system and prompt to choose one has been printed out to the command line user interface.

Project Name: Project 1: Voting System **Team# 1**

Test Stage: Unit _x_ System __ **Test Date: November 2, 2025**

Test Case ID#: TC_15 **Name(s) of Testers: Hlina**

Test Description: Test that verifies that output of HelpWindow ExplainShuffle method prints out the correct message to the screen

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes

Results: x Passed

Preconditions for Test:
Junit 5 testframe work available

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1					
2	HelpWindow explainShuffle method is called to output explanation of what shuffle is used for in the system and prompts user to select if they want it on or off.	HelpWindow Class	HelpWindow.explainShuffle()	HelpWindow.explainShuffle()	Pass
3					
4					

Post condition(s) for Test:
Explanation of shuffle mode has been printed out to the command line user interface.

Project Name: Project 1: Voting System **Team# 1**

Test Stage: Unit _x_ System __

Test Date: November 2, 2025

Test Case ID#: TC_15

Name(s) of Testers: Hlina

Test Description: Test that verifies that output of HelpWindow ExplainSeats method prints out the correct message to the screen

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes___ ___

Results: x_____ Passed

Preconditions for Test:
Junit 5 testframe work available

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1					
2	HelpWindow explainSeats method is called to output explanation of why seats are needed and how to enter them into the system.	HelpWindow Class	HelpWindow.explainSeats()	HelpWindow.explainSeats()	Pass
3					
4					

Post condition(s) for Test:
Explanation of how to enter seats/number of expected winners into the system has been printed out to the command line user interface.

Project Name: Project 1: Voting System **Team# 1**

Test Stage: Unit _x_ System __

Test Date: November 2, 2025

Test Case ID#: TC_15

Name(s) of Testers: Hlina

Test Description: Test that verifies that output of HelpWindow ExplainAudit method prints out the correct message to the screen

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes___ ___

Results: x_____ Passed

Preconditions for Test:
Junit 5 testframe work available

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1					
2	HelpWindow explainAudit method is called to output explanation of where audit file can be found.	HelpWindow Class	HelpWindow.explainAudit()	HelpWindow.explainAudit()	Pass
3					
4					

Post condition(s) for Test:
Explanation of where the audit file can be found has been printed out to the command line user interface.

Project Name: Project 1: Voting System **Team# 1**

Test Date: **November 2, 2025**

Test Stage: Unit _x_ System __

Test Case ID#: TC_16

Name(s) of Testers: Arish Soomar

Test Description: Verify that the Plurality constructor initializes with election seats, candidates, and ballots.

Indicate where are you storing the tests (what file) and the name of the method/functions being used.
Verify that the Plurality constructor properly initializes an election with seats, candidates, and ballots.

Automated: yes

Results: x Passed

Preconditions for Test:
JUnit 5 test framework available, Plurality class compiled.

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Create 2 candidates (Alice-Party A, Bob-Party B)	Candidate IDs: 0, 1	Candidates created successfully	Candidates created	
2	Create Plurality election with 1 seat, 2 candidates, empty ballots	totalSeats=1, candidates list, empty ballots list	Election object created	Election created	
3	Verify totalSeats equals 1	getTotalSeats()	Returns 1	Returns 1	
4	Verify availableSeats equals 1	getAvailableSeats()	Returns 1	Returns 1	
5	Verify candidates size is 2	getCandidates().size()	Returns 2	Returns 2	
6	Verify winners list is empty	getWinners().isEmpty()	Returns true	Returns true	
7	Verify losers list is empty	getLosers().isEmpty()	Returns true	Returns true	

Post condition(s) for Test:
Plurality election object properly initialized with expected values. No election has been run yet.

Project Name: Project 1: Voting System **Team# 1**

Test Date: **November 2, 2025**

Test Stage: Unit _x_ System __

Test Case ID#: TC_17
Test Description: Testing tie cutoff.

Name(s) of Testers: Arish Soomar

Indicate where are you storing the tests (what file) and the name of the method/functions being used:
`testTieAtCutoff()`

Automated: yes

Results: x Passed

Preconditions for Test:
4 candidates available
2 seats to be filled
Alice leads with 10 votes
Bob and Charlie tied with 5 votes each

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Create 4 candidates	Alice, Bob, Charlie, Diana	4 candidates created	Candidates created	
2	Create 10 ballots for Alice	Ballot IDs 1-10, preference: [0]	10 ballots created	10 ballots created	
3	Create 5 ballots for Bob	Ballot IDs 11-18, preference=[1]	5 ballots created	5 ballots created	
4	Create 5 ballots for Charlie	Ballot IDs 19-23, preference=[2]	5 ballots created	5 ballots created	
5	Create 2 ballots for Diana	Ballot IDs 24-25, preference=[3]	2 ballots created	2 ballots created	
6	Initialize election	totalSeats=2, 4 candidates,	Election created	Election created	

		25 ballots			
7	Run the election	runElection()	Election completes	Completed	
8	Verify winner count	getWinners().size()	Election complets	Election completes	
9	Verify first winner	N/A	2 winners	2 winners	
10	Verify second winner	N/A	Alice	Alice	
11	Verify loser count	N/A()	Bob or Charlie	Bob or Charlie	
12	Verify election is finished	N/A	2 losers	2 losers	

Post condition(s) for Test:
Alice wins first seat and one of Bob/Charlie wins second seat, tie breaking happens then there are 2 losers.

Project Name: Project 1: Voting System

Team# 1

Test Date: **November 2, 2025**

Test Stage: Unit _x_ System __

Test Case ID#: TC_18
Test Description: Unit test for the breakTie() method to verify it returns a valid index when given a list of tied candidates.

Name(s) of Testers: Arish Soomar

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

`testBreakTie()`

Automated: yes

Results: x

Preconditions for Test:
3 candidates exist
Plurality election object created
List of tied candidates prepared

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Create 3 candidates	Alice, Bob, Charlie	3 candidates created	3 candidates created	
2	Initialize election	t1 seat, 3 candidates, empty ballots	Election object created	Election created	
3	Create tied candidate list	ArrayList with 3 candidates	List created	List created	
4	Add Alice to tied list	candidates.get(0)	Alice added	Alice added	
5	Add Bob to tied list	candidates.get(1)	Bob added	Bob added	
6	Add Charlie to tied list	candidates.get(2)	Charlie added	Charlie added	
7	Verify index validity	NA	index >= 0 && index < 3	Valid index confirmed	

Post condition(s) for Test:
Valid random index returned.

Project Name: Project 1: Voting System **Team# 1**

Test Date: **November 2, 2025**

Test Stage: Unit _x_ System __

Test Case ID#: TC_19
Test Description: Verify that isFinished() correctly reports election status before and after running election.

Name(s) of Testers: Arish Soomar

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes

testIsFinished()

Results: x _____ **Passed** _____

Preconditions for Test:
2 candidates available
1 ballot for Alice

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Create 2 candidates	Alice, Bob	2 candidates created	2 candidates created	
2	Create 1 ballot	Ballot ID = 1 pref = 0 for allice	1 ballot created	1 ballot created	
3	Initialize election	1 seat, 2 candidates, 1 ballot	Election object created	Election object created	
4	Check isFinished()	N/A	N/A	Returns false	
5	Verify avail seats	N/A	1	Should = 1	
6	Run election	N/A	Election executes	Election completes	
7	CheckisFinished()	N/A	True	True	
8	Verify available seats	N/A	Get avail = 0	Get avail = 0	

Post condition(s) for Test:
Election marked as finished; available seats = 0; Alice is winner; Bob is loser!!

Project Name: Project 1: Voting System **Team# 1**

Test Date: **November 2, 2025**

Test Stage: Unit _x_ System __

Test Case ID#: TC_20
Test Description: Test clear winner

Name(s) of Testers: Arish Soomar

Indicate where are you storing the tests (what file) and the name of the method/functions being used.
testCSVClearWinner()

Automated: yes___ ___

Results: x_____ Passed _____

Preconditions for Test:
CSV file
Test_clear_winner.csv exists in ElectionSystem directory
File format: First row = candidate names, subsequent rows = ballots with "1" marking votes

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Load CSV	test_clearwinner.csv	File loaded	File loaded	
2	Create File handler	New FileHandler	Handler	Handler	
3	Read file	fh.readfile()	File parsed	File parsed	
4	Get candidate name	fh.getCandidnate	Reutnrs Arish Hlina rahim	Returns 3 names	
5	Create candidate object	Id 0-2, names from csv	3 candidate objects	3 objects created	
6	Get ballot data	fh.getballots()	Returns ballots row	Ballot retreived	
7	Verify winner count	N/A	1 winner	1Winner	

Post condition(s) for Test:
Rahim declared winner with 5 votes; Arish (2 votes) and Hlina (1 vote) are losers; CSV data successfully integrated.

Project Name: Project 1: Voting System **Team# 1**

Test Date: November 2, 2025

Test Stage: Unit _x_ System __

Test Case ID#: TC_21

Name(s) of Testers: Arish Soomar

Test Description: Integration test for 3-way tie scenario loaded from CSV file (Rahim, Arish, Hlina each with 3 votes).

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

testCSVTieScenario()

Automated: yes___ ___

Results: x

Preconditions for Test:
CSV file
test_tie_scenario.csv
exists
File contains 10 ballots: 3 for Rahim, 3 for Arish, 3 for Hlina, 1 for Adam

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Load CSV	test_clearwinner.csv	File loaded	File loaded	
2	Create File handler	New FileHandler	Handler	Handler	
3	Read file	fh.readfile()	File parsed	File parsed	
4	Get candidate name	fh.getCandidnate	Returns Arish Halina Rahim	Returns 3 names	
5	Create candidate object	Id 0-2, names from csv	3 candidate objects	3 objects created	
6	Get ballot data	fh.getballots()	Returns ballots row	Ballot retrieved	
7	Parse data	Convert to ballot	10 ballots	10 ballots created	

8	Initialize election	1 seat, 4 candidates, 10 ballots	Election object	Election object created	
9	Run Election	runElection()	Election completes	Election completes	
10	Verify winner count	N/A	1 winner	1Winner	

Post condition(s) for Test:
One of Rahim/Arish/Hlina randomly selected as winner with 3 votes. The remaining two plus Adam become losers and tie properly broken.

Project Name: Project 1: Voting System **Team# 1**

Test Date: **November 2, 2025**

Test Stage: Unit _x_ System __

Test Case ID#: TC_22
Test Description: Integration test for multi-seat (3 seats) plurality election loaded from CSV file.

Name(s) of Testers: Arish Soomar

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

testCSVMultiSeat()

Automated: yes

Results: x_____ _____

Preconditions for Test:
CSV file
Test_multi_seat.csv exists
File contains 19 ballots distributed among 5 candidates
3 seats available

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1	Load CSV file	test_multi_seat.csv	Candidates created successfully	File loaded	
2	Create FileHandler	new FileHandler()	Handler created	Handler created	
3	Read file	fh.readfile()	File parsed	File parsed	
4	Get candidate name	fh.getCandidnate	Returns Arish Halina Rahim	Returns 5 names	
5	Create candidate object	Id 0-4, names from csv	5 candidate objects	5 objects created	
6	Get ballot data	fh.getballots()	Returns ballots row	Ballot retrieved	
7	Parse data	Convert to ballot	19 ballots	19 ballots created	
8	Initialize election	1 seat, 4 candidates, 10 ballots	Election object	Election object created	
9	Run Election	runElection()	Election completes	Election completes	
10	Verify winner count	N/A	3 winners and 2 losers	3 winners and 2 losers	

Post condition(s) for Test:

Top 3 vote-getters (Rahim, Arish, Hlina) win seats; Adam and Sam become losers; all 3 seats filled.

Project Name: Project 1: Voting System **Team# 1**

Test Stage: Unit x_ System __

Test Date: 11/1/2025

Test Case ID#: TC_23

Name(s) of Testers: Rahim Bardai

Test Description:

Test to see if a non-existent file is going to throw and IO exception

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes x__ no ___

Results: Pass x Fail

Preconditions for Test: FileHandler exists, getNumVotes(), getCandidates(), and getBallots() exists, no prior state of data required.

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1					
2	Setup non-exisistent tempFile	tempFile	File is non-existent	File is non-existent	Pass
3	Check if thrown exception	tempFile	IOException	IOException	Pass
4					

Post condition(s) for Test:

Project Name: Project 1: Voting System **Team# 1**

Test Stage: Unit x_ System __

Test Date: 11/1/2025

Test Case ID#: TC_24

Name(s) of Testers: Rahim Bardai

Test Description:

Test to see if an empty file still passes through

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes_x___ no ___

Results: Pass __x___ Fail_____

Preconditions for Test: FileHandler exists, getNumVotes(), getCandidates(), and getBallots() exists, no prior state of data required.

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1					
2	Setup empty tempFile	tempFile	File is empty	File is empty	Pass
3	Check if num votes is empty	getNumVotes()	0	0	Pass
4	Check if get candidates is empty	getCandidates().empty()	True	True	pass
	Test if get ballots is empty	getBallots().empty()	True	True	Pass

Post condition(s) for Test:

Project Name: Project 1: Voting System **Team# 1**

Test Stage: Unit _ System _x_ **Test Date: 11/3/2025**

Test Case ID#: TC_25 **Name(s) of Testers: Rahim Bardai**

Test Description:

Test if the full election system main works for Plurality

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes_x_ no ___

Results: Pass __x___ Fail_____

Preconditions for Test: All Java files exist, and subfile are created (Main, Election, STV, Plurality, FileHandler, Ballot, Candidate).

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1			`String userInput = "ElectionSystem/test/test_clear_winner.csv\n" + "plurality\n" + "no\n" + "1\n"; simulateInput(userInput);`	`String userInput = "ElectionSystem/test/test_clear_winner.csv\n" + "plurality\n" + "no\n" + "1\n"; simulateInput(userInput);`	Pass
2	Simulate User Input	User Input			

3	Check print Status	Terminal Print Messages	Welcome to the Election System Plurality Election Results Total Seats: 1 Winners: Rahim 5 votes	Welcome to the Election System Plurality Election Results Total Seats: 1 Winners: Rahim 5 votes	Pass
4					

Post condition(s) for Test:

Project Name: Project 1: Voting System **Team# 1**

Test Stage: Unit _ System _x_ **Test Date: 11/3/2025**

Test Case ID#: TC_26 **Name(s) of Testers: Rahim Bardai**

Test Description:

Test if the full election system main works for STV with shuffle disabled

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes x no ___

Results: Pass x Fail

Preconditions for Test: All Java files exist, and subfile are created (Main, Election, STV, Plurality, FileHandler, Ballot, Candidate).

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1					
2	Simulate User Input	User Input	"ElectionSystem/test/test_multi_seat.csv\n" + "stv\n" + "yes\n" + "2\n";	"ElectionSystem/test/test_multi_seat.csv\n" + "stv\n" + "yes\n" + "2\n";	Pass
3	Check print Status	Terminal Print Messages	Welcome to the Election System STV Election Results Total Seats: 2 Winners: Rahim Arish	Welcome to the Election System STV Election Results Total Seats: 1 Winners: Rahim Arish	Pass

4					

Post condition(s) for Test:

Project Name: Project 1: Voting System **Team# 1**

Test Stage: Unit _ System _x_ **Test Date: 11/3/2025**

Test Case ID#: TC_27 **Name(s) of Testers: Rahim Bardai**

Test Description:

Test if the full election system main works for STV without shuffle disabled

Indicate where are you storing the tests (what file) and the name of the method/functions being used.

Automated: yes_x__ no ___

Results: Pass __x_____ Fail _____

Preconditions for Test: All Java files exist, and subfile are created (Main, Election, STV, Plurality, FileHandler, Ballot, Candidate).

Step #	Test Step Description	Test Data	Expected Result	Actual Result	Notes
1					
2	Simulate User Input	User Input	"ElectionSystem/test/test_multi_seat.csv\n" + "stv\n" + "no\n" + "2\n";	"ElectionSystem/test/test_multi_seat.csv\n" + "stv\n" + "no\n" + "2\n";	Pass
3	Check print Status	Terminal Print Messages	Welcome to the Election System STV Election Results Total Seats: 2 Winners: Rahim Arish	Welcome to the Election System STV Election Results Total Seats: 1 Winners: Rahim Arish	Pass
4					

Post condition(s) for Test: